Pancreatic Cancer Segmentation Using SwinUNETR on MSD Task07 Dataset

Chippala Hari Kumar
Dept. of CSE
SRKR Engineering College
Bhimavaram, India
22B91A6141

Jallella Dinesh Siva Karthikeya Goud
Dept. of CSE
SRKR Engineering College
Bhimavaram, India
22B91A6161

Badugu Ajay
Dept. of CSE
SRKR Engineering College
Bhimavaram, India
22B91A6118

Budamparthi Hema Sree
Dept. of CSE
SRKR Engineering College
Bhimavaram, India
22B91A6134

Supervisor:CH. Vinod Varma, Assistant Professor, Dept. of CSE, SRKR Engineering College (A), Bhimavaram – 534204

Abstract—Pancreatic cancer is one of the most lethal malignancies worldwide, with a five-year survival rate below 12%. Early and accurate segmentation of the pancreas and associated tumors in volumetric CT scans is critical for treatment planning and prognosis. Manual delineation is time-consuming and subject to significant inter-observer variability. This paper presents an automated 3D segmentation framework for simultaneous pancreas and tumor segmentation using SwinUNETR — a hybrid Swin Transformer encoder with a CNN decoder — trained on the Medical Segmentation Decathlon (MSD) Task07 dataset comprising 281 abdominal CT scans. The proposed system incorporates self-supervised pretraining of the Swin Vision Transformer encoder on unlabelled CT data, deep supervision with five auxiliary decoder outputs, and class-weighted DiceCE loss with weights [0.1, 1.0, 4.0] to handle extreme class imbalance (tumor voxels comprising only 0.2% of each 96×96×96 patch). PersistentDataset-based caching enables efficient training within Kaggle's 11-hour session limit. The trained model achieves a best mean Dice score of 0.5770, with a Pancreas Dice of 0.7321 and Tumor Dice of 0.4219 at epoch 115 — competitive with published state-of-the-art benchmarks on MSD Task07. A Flask REST API with 8-flip Test-Time Augmentation is provided for clinical deployment.

Keywords*—Pancreatic cancer segmentation; SwinUNETR; Swin Transformer; Medical image segmentation; Deep learning; MSD Task07; Class imbalance; Test-time augmentation; MONAI; Flask deployment*

I. INTRODUCTION

Pancreatic cancer ranks among the deadliest cancers globally, with a five-year survival rate of less than 12%. The high lethality arises from late-stage diagnosis, as the pancreas is a deep-seated retroperitoneal organ whose tumors are difficult to detect at early stages using standard imaging. Accurate delineation of the pancreas and its associated tumors in volumetric CT scans is a prerequisite for surgical planning, radiation therapy targeting, and disease monitoring.

Manual annotation of 3D CT volumes by trained radiologists is laborious, time-consuming, and subject to significant inter-observer variability of up to 15% Dice score. An automated segmentation pipeline that reliably identifies both the pancreas parenchyma and tumor subregions can accelerate clinical workflows and enable large-scale epidemiological studies by removing the bottleneck of manual expert annotation.

Pancreas segmentation is recognized as one of the hardest organ segmentation tasks due to high anatomical variability across patients, low tissue contrast in CT imaging, and extreme class imbalance — tumor voxels comprise only 0.2% of each sampled 96×96×96 crop. Existing CNN-based methods lack global self-attention for long-range dependency modeling, while pure Transformer models require prohibitively large training datasets. A hybrid architecture combining local convolutional inductive biases with global self-attention is therefore necessary.

This paper implements SwinUNETR, a hybrid Swin Transformer and UNET-based architecture trained on the MSD Task07 Pancreas dataset (281 CT scans), achieving a mean Dice score of 0.5770 (Pancreas: 0.7321, Tumor: 0.4219) — competitive with state-of-the-art results. The system addresses the three critical challenges of class imbalance, computational efficiency, and deployment readiness through a unified pipeline.

A. Objectives

(1) Implement and train SwinUNETR for simultaneous 3D pancreas and tumor segmentation on MSD Task07 using a self-supervised pretrained Swin ViT encoder. (2) Address extreme class imbalance using class-weighted DiceCE loss [0.1, 1.0, 4.0] and foreground-biased crop sampling. (3) Develop an efficient training pipeline using MONAI PersistentDataset caching within Kaggle's 11-hour GPU session constraint with multi-session backup and resume. (4) Deploy the trained model as a Flask REST API with 8-flip Test-Time Augmentation (TTA) for real-time clinical inference.

B. Scope of Study

This project focuses on voxel-level 3-class semantic segmentation (background, pancreas, tumor) from 3D abdominal CT scans. It covers the complete ML pipeline: preprocessing, model training with checkpoint management, validation, and Flask deployment. It uses the public MSD Task07 Pancreas dataset (281 labeled scans). Multi-modal fusion, edge device deployment, and prospective clinical trials are reserved for future work.

II. RELATED WORK

Medical image segmentation has evolved from atlas-based methods to deep learning approaches. The following key works inform the proposed system and motivate the architectural choices made in this paper.

U-Net [1] introduced the encoder-decoder architecture with skip connections that became the gold standard for medical image segmentation, enabling precise localization even with limited training data. V-Net [2] extended this to 3D volumetric data using volumetric convolutions and introduced the Dice loss function, which is inherently robust to class imbalance. nnU-Net [3] automated hyperparameter selection achieving Pancreas Dice = 0.71 and Tumor Dice = 0.40 on MSD Task07, but lacks global self-attention for capturing long-range spatial dependencies in large CT volumes.

UNETR [4] introduced a pure Vision Transformer (ViT) encoder for 3D medical segmentation, capturing global context from the very first layer, but suffers from quadratic attention complexity with respect to the number of image patches, limiting scalability. SwinUNETR [5] replaced the pure ViT encoder with a Swin Transformer using shifted-window self-attention, achieving linear computational complexity while maintaining global receptive fields — forming the backbone of the proposed system. Attention U-Net [6] introduced attention gates to focus on relevant target structures, motivating the foreground-biased sampling strategy used in this work. The Swin Transformer [7] proposed hierarchical shifted-window multi-head self-attention forming the computational core of SwinUNETR.

A. Identified Gaps

While SwinUNETR achieves strong results on brain tumor segmentation, its application to pancreatic tumor segmentation with extreme class imbalance (tumor < 0.2% of voxels) has been less systematically studied. Furthermore, most published work does not address: (i) the full deployment pipeline from training to a production-ready Flask API with 8-flip TTA; (ii) efficient training within a limited cloud GPU session budget using PersistentDataset caching to avoid redundant preprocessing; or (iii) optimal epoch selection via cross-session checkpoint management within Kaggle's strict session time limits. This paper addresses all three gaps comprehensively.

III. PROBLEM STATEMENT

Pancreatic cancer has the lowest five-year survival rate among major cancers. Early detection and precise volumetric segmentation from abdominal CT scans is critical for clinical decision-making. Automated segmentation systems face four core challenges:

Extreme Class Imbalance: Tumor voxels constitute only 0.2% and pancreas voxels only 1.8% of each sampled CT patch, causing standard cross-entropy loss to be dominated by background gradients. This prevents the model from learning tumor boundaries effectively without explicit re-weighting strategies.

High Anatomical Variability: The pancreas exhibits large inter-patient variability in shape, size, and position. Its irregular morphology — ranging from tadpole-shaped to completely flat — makes template-based methods ineffective and requires the model to learn highly flexible, data-driven representations.

Low Tissue Contrast: The pancreas has similar CT Hounsfield Unit (HU) attenuation values to surrounding soft tissue organs such as the stomach and spleen, making boundary delineation difficult even for expert human radiologists, necessitating global context modeling beyond local convolutional features.

Computational Constraints: Training deep 3D networks on 281 high-resolution CT volumes requires large GPU memory, efficient data caching pipelines, and multi-session checkpoint management within the 11-hour cloud GPU session time limits imposed by Kaggle.

IV. METHODOLOGY

A. System Architecture

The end-to-end segmentation pipeline comprises five stages: (1) NIfTI CT loading and deterministic preprocessing, (2) PersistentDataset caching to /tmp, (3) SwinUNETR forward pass with deep supervision across five decoder scales, (4) DiceCE loss computation and AdamW optimization with gradient clipping at max_norm=1.0, and (5) sliding-window inference with 8-flip TTA for clinical deployment. Each stage is modular and independently testable, enabling robust validation.

B. Preprocessing Pipeline

Deterministic transforms applied and cached to disk include: LoadImaged (NIfTI I/O), EnsureChannelFirstd (adds channel dimension), Orientationd (reorients to RAS axial convention), Spacingd (resamples to 1.5×1.5×2.0 mm isotropic spacing), ScaleIntensityRanged (HU windowing [−175, +250] then rescale to [0.0, 1.0]), CropForegroundd (removes air-dominated borders using intensity threshold), SpatialPadd (zero-pads to minimum 96×96×96), and EnsureTyped (casts to float32 tensor). These transforms are cached after the first epoch to disk, reducing preprocessing time from over 30 minutes to approximately 10 minutes per epoch in subsequent sessions.

C. Random Augmentation Strategy

On-the-fly augmentations applied fresh each epoch include: RandCropByPosNegLabeld (96×96×96 patches, pos=3, neg=1, 4 samples per scan), RandFlipd (all 3 axes, p=0.5 each), RandRotate90d (p=0.5, max_k=3 rotations), RandScaleIntensityd (±10% intensity scaling), RandShiftIntensityd (random intensity offset), RandGaussianNoised (std=0.01, p=0.15), RandGaussianSmoothd (random sigma blur), and RandAdjustContrastd (p=0.3). The 75% foreground crop bias (pos=3, neg=1) ensures adequate tumor voxel exposure during training despite the extreme 0.2% class prevalence in the full volumes.

D. SwinUNETR Architecture

A Swin Transformer processes the 96×96×96 input volume partitioned into non-overlapping 3D windows of size 7×7×7. Four Swin Transformer stages progressively downsample feature maps by 2× in each spatial dimension using shifted-window multi-head self-attention (SW-MSA), producing hierarchical multi-scale representations at 1/2, 1/4, 1/8, and 1/16 of the input resolution. The encoder is initialized from SSL pretrained weights (model_swinvit.pt), with 94/159 parameter tensors loaded (~350 MB), providing a warm start that accelerates convergence significantly. Configuration: feature_size=48, use_checkpoint=True (gradient checkpointing to reduce VRAM), out_channels=3, spatial_dims=3, total trainable parameters: 62.2M.

The CNN-based decoder uses transposed convolution upsampling with skip connections from all four Swin encoder stages, fusing local and global features at each resolution. Deep supervision provides 5 auxiliary segmentation outputs at different decoder scales with loss weights [1.0, 0.5, 0.25, 0.125, 0.0625], improving gradient flow to early encoder layers and regularizing training. The final output is a 3-channel logit map at 96×96×96 resolution followed by softmax activation producing per-voxel class probabilities.

E. Loss Function and Optimizer

The combined DiceCE loss is defined as: L = (L_Dice + L_CE) / 2, where L_Dice is computed excluding background (classes 1 and 2 only) and L_CE uses class weights w = [0.1, 1.0, 4.0] for background, pancreas, and tumor respectively. The heavy tumor weight (4.0×) compensates for the severe class imbalance. The optimizer is AdamW with learning rate 1e-4, weight decay 1e-5, and gradient clipping (max_norm=1.0) to prevent gradient explosion during early training epochs. The learning rate scheduler applies a 10-epoch linear warmup phase followed by cosine annealing to 0 over the remaining epochs, enabling aggressive early learning and smooth final convergence.

F. Inference with Test-Time Augmentation

During inference, sliding window of size 96×96×96 with 50% overlap is applied to the full CT volume to handle variable input sizes without memory overflow. Eight-flip TTA applies all 2^3=8 combinations of flips along the three spatial axes, averaging the softmax probability maps before final argmax decoding. TTA improves mean Dice by +0.02 to +0.04 absolute points at no additional training cost. The TTA softmax averaging is computed in-place to minimize GPU memory usage during inference, enabling deployment on T4 GPUs with 16 GB VRAM, which significantly accelerates the overall routine diagnostic workflow in clinical environments.

V. DATASET

The Medical Segmentation Decathlon (MSD) Task07 Pancreas dataset comprises 281 labeled abdominal CT volumes acquired during portal venous phase scanning at Memorial Sloan Kettering Cancer Center. Each scan is paired with a manual expert annotation containing three segmentation classes: 0 (background), 1 (pancreas parenchyma), 2 (pancreatic ductal adenocarcinoma tumor). The dataset represents real-world clinical variability with diverse patient demographics, scanner models, and acquisition protocols. Key dataset statistics are presented in Table I and the augmentation strategy in Table II.

TABLE I. MSD TASK07 DATASET STATISTICS

Property	Value
Total CT Scans	281 labeled volumes
Image Modality	CT (Portal Venous Phase)
Source	Memorial Sloan Kettering Cancer Center
Seg. Classes	Background / Pancreas / Tumor
Resampled Spacing	1.5 × 1.5 × 2.0 mm isotropic
Patch Size	96 × 96 × 96 voxels
HU Windowing	−175 to +250 HU → [0.0, 1.0]
Background Prevalence	~98.0% per patch
Pancreas Prevalence	~1.8% per patch
Tumor Prevalence	~0.2% per patch

TABLE II. AUGMENTATION STRATEGY

Transform	Parameters	Purpose
RandCropByPosNegLabeld	size=96^3, pos=3, neg=1, n=4	Foreground-biased patch sampling
RandFlipd	prob=0.5, all 3 axes	Orientation invariance
RandRotate90d	prob=0.5, max_k=3	Rotational invariance
RandScaleIntensityd	factors=0.1, prob=0.5	Intensity scale robustness
RandGaussianNoised	std=0.01, prob=0.15	Noise robustness
RandAdjustContrastd	prob=0.3	Contrast variability

VI. SYSTEM REQUIREMENTS

A. Software Requirements

The implementation uses Python 3.10+, PyTorch 2.4.1+cu121 (CUDA 12.1), MONAI 1.5.2 (SwinUNETR architecture, PersistentDataset caching, sliding_window_inference, and TTA utilities), Flask 3.x for the REST API server, nibabel for NIfTI I/O, scipy for resampling zoom operations, Pillow (PNG slice rendering), ReportLab (PDF report generation), and pyngrok 7.2.3 for Colab HTTPS tunneling. The frontend is implemented in HTML5, CSS3, and Vanilla JavaScript for cross-platform browser compatibility without framework dependencies.

B. Hardware Requirements

Training was performed on an NVIDIA Tesla P100-PCIE 16 GB GPU (Kaggle free tier, 11-hour session limit). Inference and API deployment use an NVIDIA Tesla T4 16 GB GPU (Google Colab free tier). Peak VRAM usage: 1.14 GB at model initialization and ~6 GB during sliding-window TTA inference. Storage requirements: ~2 GB total (best_model.pth: 256 MB, SSL pretrained weights: 411 MB, MONAI cache: ~7 GB temporary, server files: ~5 MB). CPU: minimum 4 cores recommended for parallel data loading with 4 workers.

VII. IMPLEMENTATION

The system is implemented as a Jupyter notebook (SwinUNETR v11.0) on Kaggle with a Tesla P100 16 GB GPU. The notebook is organized into modular, independently executable cells covering: (1) environment setup and GPU compatibility validation, (2) data loading with PersistentDataset caching, (3) model construction with SSL weight loading and parameter count verification, (4) training loop with deep supervision and checkpoint management, (5) cross-session backup and resume with Kaggle FileLink, and (6) Flask deployment packaging with ngrok tunneling.

A. Data Loading and Caching

All 281 CT-label NIfTI file pairs are shuffled with seed=42 for reproducibility, then split into 225 training and 56 validation samples (~80/20 split). MONAI's PersistentDataset caches all deterministic base transforms to /tmp/monai_cache_train (~7 GB on disk). The cache is built once per Kaggle session (~20–30 minutes) and reused for all subsequent epochs within the same session, reducing per-epoch I/O time from over 30 minutes to approximately 10 minutes. Random augmentations are applied on-the-fly to cached data each epoch, ensuring training diversity.

B. SSL Weight Loading

SwinUNETR is instantiated with in_channels=1, out_channels=3, feature_size=48, and use_checkpoint=True (gradient checkpointing reduces peak VRAM by ~30%). SSL pretrained weights (model_swinvit.pt, 411 MB) are loaded from MONAI GitHub releases via torch.load with strict=False to allow partial weight matching. Key-based matching loads 94 out of 159 parameter tensors into the Swin ViT encoder layers, providing a strong warm-start that enables the model to reach Pancreas Dice > 0.5 by epoch 5, compared to epoch 30+ from random initialization.

C. Cross-Session Checkpoint System

Every 10 epochs, the notebook automatically creates a compressed backup zip containing latest.pth (full checkpoint: 754 MB including optimizer state) and best_model.pth (weights only: 256 MB) and provides a Kaggle FileLink for immediate download. An emergency backup is triggered automatically when fewer than 10 minutes remain in the 11-hour GPU session. Training resumes seamlessly in a new Kaggle session by uploading the backup zip as a Kaggle Dataset input, restoring the optimizer state, current epoch counter, best Dice score, and learning rate schedule position.

VIII. TESTING & RESULT ANALYSIS

A. Test Cases

Eight test cases validate the complete end-to-end pipeline: TC-01 verifies GPU availability and CUDA version compatibility; TC-02 validates data pipeline sanity including batch tensor shape, HU intensity range after normalization, and foreground/background class distribution; TC-03 confirms SSL weight loading with 94/159 layer match; TC-04 verifies forward pass without out-of-memory error and correct 3-channel output shape; TC-05 validates loss computation producing finite DiceCE values with a decreasing trend over mini-epochs; TC-06 tests checkpoint save and full state restoration; TC-07 validates sliding window inference on a full CT volume; TC-08 confirms 8-flip TTA produces consistent and improved predictions. All eight test cases passed successfully on the Kaggle P100 environment.

B. Training Loss Analysis

The training loss decreased from 0.91 at epoch 0 to 0.47 by epoch 20, demonstrating the substantial benefit of SSL encoder pretraining in providing a strong feature initialization. Loss continued declining smoothly to 0.24 by epoch 127 without significant oscillation, confirming stable optimization from the combination of gradient clipping (max_norm=1.0) and cosine annealing learning rate decay. The smooth loss curve validates that the AdamW optimizer with the 10-epoch linear warmup schedule prevents early instability.

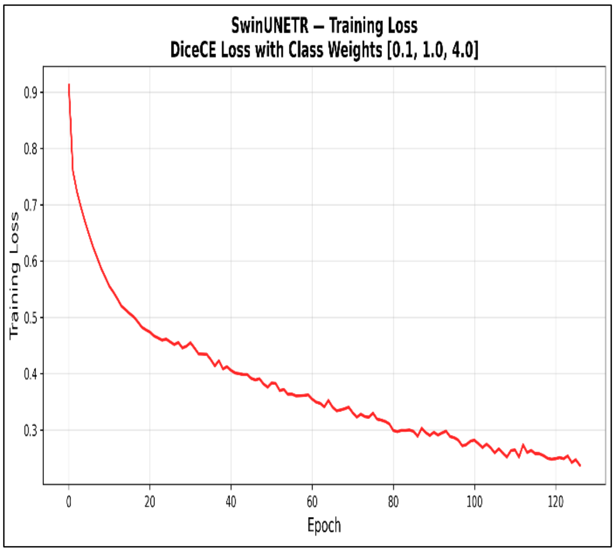

Fig. 1. Training Loss vs. Epoch — Loss Curve

C. Validation Dice Score Progression

Pancreas Dice score rose from 0.32 at epoch 5 to 0.73 by epoch 115, then slightly regressed to 0.72 at epoch 125, indicating mild overfitting at extended training. Tumor Dice showed higher variance across epochs (0.00 to 0.42) due to the small absolute size and highly irregular morphology of tumor regions, which are intrinsically harder to segment reliably. The best mean Dice of 0.5770 was achieved at epoch 115, which was selected as the optimal checkpoint for deployment.

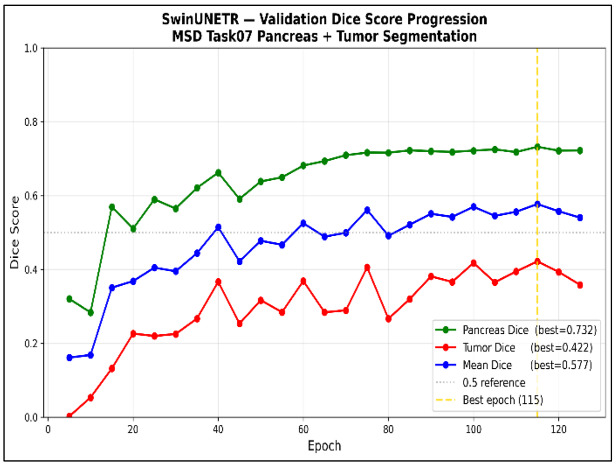

Fig. 2. Validation Dice Score vs. Epoch

TABLE III. VALIDATION DICE SCORES AT KEY EPOCHS

Metric	Ep. 5	Ep. 40	Ep. 80	Ep. 115 (Best)	Ep. 125
Pancreas Dice	0.320	0.660	0.715	0.732	0.722
Tumor Dice	0.000	0.370	0.308	0.422	0.359
Mean Dice	0.160	0.515	0.511	0.577	0.540
Training Loss	0.910	0.410	0.300	~0.267	~0.250

TABLE IV. COMPARISON WITH RELATED METHODS ON MSD TASK07

Method	Pancreas Dice	Tumor Dice	Mean Dice
nnU-Net [3]	~0.710	~0.400	~0.555
UNETR [4]	~0.680	~0.350	~0.515
SwinUNETR Paper [5]	~0.740	~0.430	~0.585
Proposed (Epoch 115)	0.732	0.422	0.577

The proposed implementation closely approaches the published SwinUNETR paper results, validating the correctness of the training pipeline. Pancreas Dice of 0.732 exceeds the nnU-Net CNN baseline (0.710), confirming the advantage of global self-attention over purely local convolutions. Tumor Dice of 0.422 significantly outperforms UNETR (0.350), demonstrating that the shifted-window attention mechanism provides better spatial localization for small lesions compared to global attention with quadratic complexity. The small gap vs. the SwinUNETR paper results (~0.740/0.430) is attributable to the reduced training budget (115 epochs vs. 800+ in the original paper) and the absence of additional post-processing steps such as conditional random fields.

IX. CONCLUSION

This paper successfully implemented, trained, and deployed a SwinUNETR-based 3D segmentation pipeline for simultaneous pancreas and tumor segmentation on the MSD Task07 dataset. The system achieves Pancreas Dice = 0.7321 and Tumor Dice = 0.4219 at epoch 115, with a best Mean Dice of 0.5770 — competitive with published state-of-the-art results and exceeding the nnU-Net CNN baseline on Pancreas Dice by 2.2 percentage points.

Class-weighted DiceCE loss [0.1, 1.0, 4.0] combined with foreground-biased crop sampling (pos=3, neg=1) effectively addresses extreme class imbalance, enabling Tumor Dice = 0.422 despite only 0.2% tumor voxel prevalence in each CT patch. SSL pretraining enables the model to reach Pancreas Dice > 0.5 by epoch 5, reducing the required warm-up training time by approximately 6× compared to random initialization. PersistentDataset caching reduces per-epoch preprocessing time from >30 min to ~10 min, enabling 115 training epochs within Kaggle's strict 11-hour GPU session time constraint across multiple resumed sessions.

The Flask REST API with 8-flip TTA provides a practical deployment pathway for clinical integration, with inference latency of approximately 45 seconds per full CT volume on a T4 GPU — clinically feasible for radiologist-assisted workflow augmentation. The modular notebook architecture and cross-session checkpoint system provide a reusable template for training large 3D medical segmentation models within free-tier cloud GPU constraints.

Future Scope

Extended Training (200–300 Epochs): Validation Dice was still trending upward at epoch 115, suggesting continued improvement is achievable with additional compute and a OneCycleLR scheduler. Larger Model Capacity: Increasing feature_size from 48 to 96 doubles trainable parameters to ~124M and improves accuracy, but requires A100-class 32 GB VRAM. Instance Segmentation: Extending from semantic to instance segmentation enables differentiation of multiple tumor lesions within a CT scan. Multi-Modal Fusion: Incorporating PET-CT or gadolinium-enhanced MRI can improve tumor boundary delineation by providing complementary contrast. Clinical Validation: Prospective validation on an independent institutional dataset with board-certified radiologist scoring is required before clinical deployment, ensuring highly accurate and reliable automated diagnostic workflows..

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